

An Empirical Study on Theories of Sentiment Analysis in Relation to Fake News Detection

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Abstract—With the speedy growth and prominence of social media, the way people interact and share information has been changed forever. This has led to an increased influence of social media platforms on people and led to rise of various issues in regards to fake news and misinformation spreading through various social media platforms. While fake news detection methodologies and techniques have advanced, fake news has also evolved to become more contextual and intent-driven, requiring a more sentiment-analytical approach. This research aims to carry out a comprehensive study of fake news detection theories, techniques, and insights in relation to sentiment analysis. It analyzes various sentiment analysis theories and fake news detection theories through studying existing research and information. This research emphasizes the importance of sentiment analysis for more effective detection of fake news.

Keywords—fake news, sentiment analysis, fake news detection, social media

I. INTRODUCTION

Fake or false news can be described as news that contains entirely or partially false information. The impact and proliferation of fake news have become a major threat to society and the integrity of information. The short and long-term negative effects on the world through the decrease in trust in mainstream media, the spread of false claims, the incitation of violence, and the growth of conspiracy theories. To circumvent fake news, several fake News detection methodologies have been developed. There have been several comprehensive studies on fake news detection alongside machine learning models trained to identify fake news through various parameters. Social media platforms have also implemented several guidelines to disrupt fake news from spreading through automated and manual methods.

However, one area in which fake news detection has not been fully explored is the sentiment analysis perspective. Sentiment analysis is a technique that can be used to identify the emotional tone of a piece of text. This can help detect fake news, as fake news articles often contain negative or emotional language. Understanding the intent of news and think pieces are as important as factual information as they can help stop the spread of fake news or hate speech in disguised patterns. Integrating sentiment-based fake news detection can help improve the effectiveness of fake news detection across social media channels and help mitigate more multidimensional types of fake news pieces.

Zhang et al. conducted an extensive study of fake news reach and dynamics. The paper tries to contextualize key fake news detection challenges, solutions, and paradigms based on current research papers. The paper also does a comparison of

the data sets used within that paper. However, the paper does not fully contextualize the difference between fake news detection sources or how important localization or region-based bias can be. It also does not make any significant improvements to the existing research [1].

Zhou et al. carries out a thorough study of fake news detection methods, theories, and frameworks. The paper does a comparative analysis and classification of several fake news detection studies along with the depth and complexity of their work. But the paper does not elaborate on non-traditional fake news sources, particularly centered around social media. The paper does not account for fake news spread in markets outside the U.S. as well. where platforms and metrics are different. It also does not discuss fake news protection or intervention in depth [2].

Zubiaga et al. conducted a comparative analysis of fake news detection metrics such as classification system, architecture, and detection approach. Using existing research and data sets. But the survey does not provide any solutions beyond those presented in its reviewed sources. The paper also lacks contextual data outside the U.S. market or social media platforms popular in emerging markets [3].

Shu et al. proposed a tri-relationship framework for fake news detection using an optimization algorithm. They require additional research because they do not extract attributes to model the intended effect of fake news from a psychological standpoint. The paper also does not explore how to identify low-quality or perhaps malicious users spreading fake news which is a key point of action in fake news detection [4].

Reis et al. tried to implement fake news supervised learning models based on fake news research and social media. It aims to present a new set of features in the model based on its performance measures of existing fake news detection model analysis and visualization but does not implement its feature set in a model for comprehensive testing or data visualization, making its proposed feature set purely hypothetical for now. The paper also does not account for fake news detection features or approaches in emerging markets or non-western regions [5].

Miguel A et al. explored the use of sentiment analysis as a tool in detecting fake news. The authors discuss how fake news can be detected by analyzing the sentiments expressed in a piece of news or related information. The paper focuses mainly on the use of sentiment analysis for detecting fake news, without discussing in-depth the potential limitations and challenges of this approach [6].

Jang et al. proposed a framework for monitoring, assessing, and forecasting the spread of erroneous data on digital channels. It used evolution tree analysis to examine the origins and evolution patterns of fake news on social media in the context of the 2016 US presidential election. The authors reveal that tweets about real news showed a wider breadth and shorter depth than tweets about fake news, indicating variations in the manner of evolution between true and false news stories. There are some gaps when it comes to establishing efficient interventions and algorithms to combat the spread of fake news on social media [7].

Ajao et al. introduced a novel approach to detect fake news by considering the relationship between sentiment and false information. The authors proposed a hypothesis that emotional phrases in text can aid in identifying fake news and rumors. By contrasting their sentiment-aware algorithm with standard text-only approaches, they demonstrate that their hypothesis is valid. The paper doesn't extensively explore the challenges of handling false positives and negatives in fake news detection, potentially leaving a gap in addressing the trade-off between accuracy and misclassification [8].

Ghanem et al. investigated the role of emotions in the detection of false information across social media and news articles. The authors' thorough research reveals that various forms of erroneous information display distinctive emotional patterns, emphasizing the crucial role that emotions play in misleading readers. The paper introduced an emotionally infused LSTM neural network model for false news detection. The study could further delve into the ethical considerations and potential biases associated with using emotional features in false information detection [9].

Shu et al. carried out an extensive evaluation of the data mining methods for identifying fraudulent information on social media. It presents the unique challenges of detecting fake news on social media and discusses existing algorithms and evaluation metrics. While the paper briefly discusses the psychological and social theories behind the creation and dissemination of fake news, it does not delve into the specific cognitive biases and social dynamics that can affect how individuals interact with and perceive fake news [10].

Most of the studies outlined here, while explore various areas of fake news detection techniques and theories, they do not explore sentiment analysis theories and their relation to fake news detection. This leaves the area of sentiment analysis and its effectiveness in fake news detection, fairly unexplored and presents a significant research gap. This paper aims to conduct an empirical study on fake news detection theories, techniques and derive key insights into fake news detection to see how much they take sentiment analysis into account. It will also discuss how sentiment analysis should and can be used to enhance existing fake news detection methods and create a more context-aware fake news detection process.

This paper is organized as follows: Section II covers the Theories and insights of fake news, Section III describes the fake news detection methods, Section IV covers the sentiment analysis in fake news detection and finally Section V concludes the paper.

II. THEORIES AND INSIGHTS OF FAKE NEWS

A. Understanding the Challenge of Fake News Definition

The concept of fraudulent news has slowly gained traction in mainstream media over the past decade. However, it is still difficult to derive a concise and universally accepted definition of fake or fraudulent news which has posed several challenges in the analysis and evaluation of fake news-related studies. Without a consistent definition the usage of related terminology to fake news such as satire, disinformation, misinformation, cherry picking, clickbait, and rumors interchangeably. However, these terms can be further distinguished based on authenticity, intent, and acceptance as a news source in the media. Fig. 1. illustrates the classifications of fake news terminology using these criteria.

Due to a lack of universal definition, there are several issues behind classifying any piece of media as explicitly false. While articles intentionally and verbally false information news are simple enough to classify as fake news it's much more difficult while taking intent into account.

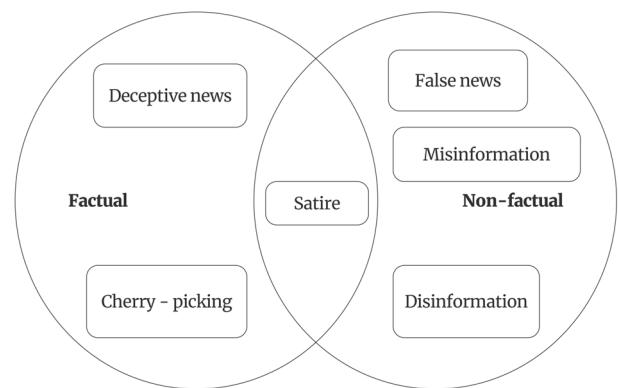


Fig. 1. Breakdown of various news items.

For example, an article disproving an original claim of a false claim might still contain the claim within. Identifying intent is key here as well as parody or satire often use misinformation or exaggeration for comedic effect. In addition, the definition of news has become harder to narrow down due to the digitization of news, which has challenged conventional definitions. Under these circumstances, a wide and narrow definition of fake news has been proposed to address the current challenges.

A further issue of broadly defining fake news as false news, which can be generated by journalists and non-journalists, comprises speeches, articles, statements, claims, and posts related to public figures and organizations. This definition only enforces entry-level constraints while weakening the requirement of information intent. A narrower definition of fake news, defines fake news as any intentionally false news published by any media outlet as it addresses public perception of false news while emphasizing authenticity and intent. It also takes the validity of the news source or outlet into account to further verify a news source. It also remains crucial to establish an inclusive definition to establish an absolute basis for deceptive news analysis as the indication of fraudulent information should take more variables into account than just news and should consider the entire information ecosystem. These variables and factors

include but are not limited to information dissemination channels, consumers, and the social and political contexts in which information is produced and consumed.

B. Sentimental Theories Regarding Fake News

With the rise in the flow of deceptive news and increase in information channels that are used to spread and consume news, stopping the propagation of false news has become increasingly challenging due to the ability to spread fast and influence people's beliefs in behavior. Mitigating fake news in such a dynamic environment requires a multidimensional approach combining various disciplines and theories. Fake news detection theory for dynamic information channels must also work to incorporate social, psychological, and economic theories regarding influence and sentiment into fake news detection. Table I. demonstrates some key theories regrading user sentiment in relation to fake news [12].

TABLE I. THEORIES REGARDING FAKE NEWS SENTIMENT

Theory name	Hypothesis
Undeutsch hypothesis	A tendency to believe the content and quality of a statement based on a factual experience is distinct from the experience itself.
Reality monitoring	A tendency to believe real-life events contain a greater amount of sensory-perceptual information
Four-factor theory	A tendency to believe that the expression of lies differs from the truth in the context of emotion, arousal, behavior control, and thinking.
Information manipulation theory	A tendency to believe that deception frequently involves an excessive amount of information
Conservatism bias	A demeanor to revise own beliefs sufficiently if or when presented with previously unseen evidence
Semmelweis reflex	An innate tendency to not accept new evidence when it deviates from established beliefs
Echo chamber effect	Sentiments are accentuated and boosted through internalized communication within an enclosed channel or space
Attention bias	Perception is affected by recurring thoughts
Validity effect	A tendency to believe the information to be correct after repeated exposure
Bandwagon effect	Believing or taking part in an activity due to the participation of peers
Normative influence theory	The influence of others leading to individuals to conform to be accepted by their peers
Social identity theory	One's self-identity derives from a perceived placement in a social group or class
Availability cascade	A tendency to adopt insights when they become popular in society

Confirmation bias	A predisposition to believe the information on the basis of it reinforcing previously held beliefs
Selective exposure	A propensity to trust the information on the basis of only being exposed to that set of beliefs
Desirability bias	A tendency to accept information when its insights a positive emotional response
Illusion of asymmetric insight	A tendency to favor subjective judgment over reliable factual information sources
Naïve realism	A tendency to favor direct awareness of senses and direct awareness of objects
Overconfidence effect	A tendency to favor subjective confidence over factual awareness
Prospect theory	A tendency to make decisions or beliefs based on the value of the decisions in terms of profit or loss
Valence effect	A tendency to believe in the greater likelihood of a positive outcome or truth over the actuality

These theories provide valuable insights for analyzing fake news, identifying its characteristics, and developing well-justified models for detection and intervention [2][11]*.

Some of the hypotheses from these theories state that fake news can be identified through differences in writing style, quality, word count, and expressed sentiment. While these theories target deceptive statements, they can be used to verify whether these attributes are practically feasible to discriminate between false information, fabricated news, and the actual facts through the use of big fake news data. This is how some of these principals can be utilized to identify fake news by identifying key characteristics in the news item. Fig. 2. below showcase some common classifications of various types of fake news based these theories.

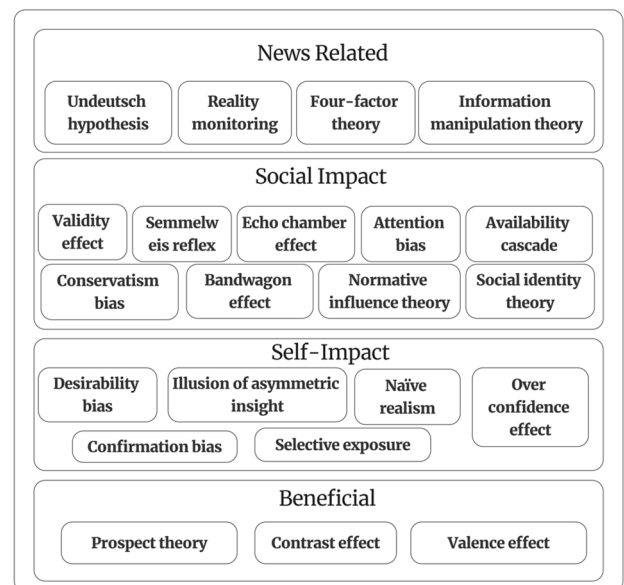


Fig. 2. Classifications of sentimental theories.

Theories centered on user-related focus on the users' traits in the false news cycle activities such as posting, liking, commenting, and forwarding. Fake news can attract both indoctrinated and normal users where indoctrinated users spread fake news due to some actual or perceived incentive. Conversely, most impartial users share fake news without recognizing any factual deficiencies and falling victim to conscious or subconscious psychological vulnerability.

Understanding the potential roles that these fundamental theories can play in fake news research requires further interdisciplinary investigations. One way these theories can improve fake news mitigation is by utilizing the foundation of these theories to identify user intent, user credibility, and sentiment. Fake news detection strategies from a user-centric perspective must take users of divergent levels of credibility and intent into account. Personal recommendations or curation of original factual news and refuted fake ones can be useful for neutral users, but not for users already inclined to share fake news due to any pre-existing incentive or bias. Quantifying self and social-impact can also optimize fake news research and improve our understanding of the psychological factors that drive fake news activities.

Besides these theories, interdisciplinary investigations that utilize theories from social sciences and economics provide a promising approach to studying fake news. Through the incorporation of these theories, it is possible to develop more effective strategies for detecting and intervening in fake news activities, and ultimately, mitigate the negative impacts of fake news on society.

III. UNDERSTANDING DETECTION METHODS OF FAKE NEWS

Detection cycle of fake news comprises of several stages, each of which is critical in the overall process. Fig. 3 demonstrates how a fake news detection model builds its knowledgebase through fact extraction and checking over multiple phases. The first stage involves data collection, where large datasets of news articles and other relevant information are gathered for analysis. These datasets are often curated and labeled to facilitate the training of machine learning models, which can then be used to label news articles as either original or fake.

The following cycle involves feature extraction, where specific features or attributes are extracted from the collected data to represent the key characteristics of fake news. The attributes or features could be related to the language used, the tone of the article, or the source of the information. These features are then fed into machine learning algorithms, which are trained to identify patterns in the data and determine it as genuine or fraudulent [12].

The third stage includes training of the machine learning models. This stage is crucial because the accuracy of the models determines the effectiveness of the entire fake news detection cycle. A wide range of algorithms may be utilized to train machine learning models. The selection of algorithm relies on the size and complexity of the dataset and the required level of accuracy [13].

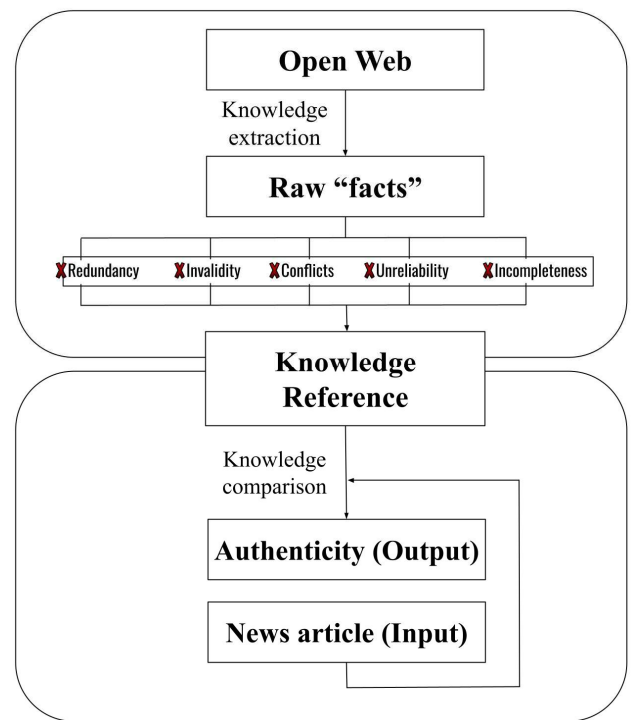


Fig. 3. Training mechanism of fake news detection system.

The fourth stage involves model evaluation, where the performance of the machine learning models is measured using metrics such as precision, accuracy, recall, and F1-score. These metrics provide insight into the model's ability to correctly classify real and fake news articles. If the model's performance is unsatisfactory, further iterations of training and evaluation are conducted until the desired level of accuracy is achieved [14].

The final stage involves model deployment, where the trained machine learning models are integrated into fake news detection systems that can be used by individuals and organizations to detect and verify the authenticity of news articles. These systems can be deployed as standalone applications or integrated into existing news platforms or social media platforms [15][16].

IV. SENTIMENT ANALYSIS FOR FAKE NEWS DETECTION

As previously mentioned, analysis of sentiment is a sub-genre of natural language processing focused on identifying and extracting emotional tone and information that can be described as subjective from the text [17]. Sentiment analysis has various existing consumer-focused applications already in use in varying capacities such as monitoring social media and analyzing consumer feedback. However, its utilization in fake news detection remains less proficient.

Sentiment analysis can be combined with more traditional approaches for fake news detection to develop new detection models that are scalable, objective, and automated and correctly identify the content of a message [18]. This is most effective in identifying the tone of content and sentiment that can help provide critical insights regarding content veracity. This can be particularly helpful in identifying sensationalized content targeted at masking false news behind opinion pieces. Here are some of the ways, sentiment analysis can be used to boost fake news detection:

A. Utilizing Sentiment Analysis to Identify Emotional Language Patterns in Fake News

One of the usages of sentiment analysis is identifying emotional content in news stories. Identifying emotional content in news stories and classifying them can help people understand the sentimental state of content and make assessments of the intent. This can help sort news content and mark potentially problematic fake news stories [19].

B. Utilizing Sentiment Analysis to Identify the Impact of False News on Users

Another way sentiment analysis can be used in deceptive news detection is by identifying the impact fake news might have on users [20]. Sentiment analysis can be used to identify emotional cues that are associated with human reactions. Understanding what emotional reaction fake news can elicit in people can help determine how specific communities react to fake news stories which can help social platforms moderate news content more effectively.

C. Utilizing Sentiment Analysis to Detect Inconsistencies in Tone

Sentiment analysis can also be used to identify tone in text-based content and tone inconsistencies [21]. Inconsistencies in tone can be used as an identifier of modified content or altered news stories with fake news or false claims being injected. This can help media channels to accurately assess news stories and ensure the authenticity of the information.

D. Utilizing Sentiment Analysis to Detect Changes in Tone over a Period

Another way sentiment analysis can be used is to detect tone changes over a certain period [22]. Understanding tone changes in content over some time can help flag news sources that become full of fake news and gradually degrade in authenticity. This can help monitor news sources over a long period and help maintain the authenticity of news.

V. CONCLUSION

The challenge of fake or fraudulent news detection has become more and more difficult due to the growth of several media consumption channels and the rapid propagation of information. Growing internet and smart device adoption have created new information and media consumption channels, such as social media. With these new channels, the ability to read and share information online has become more accessible and less centralized. This research goes over how fake news detection methods need to be more sentiment centric and highlights how, in a world where the context and intent of information are as important as factual evidence, the importance of sentiment analysis in fake news detection cannot be understated. It shows how sentiment analysis and fake news theories can be used to improve fake news detection across various channels. It also establishes that it's feasible to develop fake news detection methods that can understand context and intent at scale and improve fake news detection and moderation across a diverse range of media platforms such as context-aware mitigation methods built using sentiment analysis. Ultimately, this research showcases the importance of sentiment analysis in the fake

news detection cycle and its role in mitigating more contextually sensitive forms of fake news.

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